SPEECH EMOTION RECOGNITION USING SPECTRAL FEATURES AND DEEP LEARNING

1. INTRODUCTION

Speech Emotion Recognition (SER) aims to detect human emotions such as happiness, anger, and sadness from audio signals. This task explores two complementary approaches: handcrafted spectral features Spectral Centroid (SC), Spectral Bandwidth (SBW), and Spectral Band Energy (SBE) classified using a linear SVM, and deep-learning-based models using 1D and 2D Convolutional Neural Networks (CNNs). The experiment uses the RAVDESS dataset containing balanced emotional speech samples. Performance is evaluated in terms of accuracy to identify which method better captures emotional cues.

2. SPECTRAL FEATURE EXTRACTION

2.a Spectrogram and Mel Representation

Each audio file was pre-processed by trimming silence and applying a 25 ms Hamming window with 10 ms overlap. A spectrogram was generated to visualise temporal frequency variation. Figure 1 (spectrogram) shows distinct formant and energy distributions across emotions for example, anger exhibits stronger high-frequency energy, while sadness concentrates in lower frequencies.

Although many pipelines convert frequencies into Mel scale, the tutor clarified that the provided band edges (e.g., 300 Hz, 627 Hz, etc.) are already expressed in Mel scale. Therefore, the use of librosa.mel_frequencies was unnecessary and removed.

2.b Spectral Centroid (SC)

SC represents the "centre of mass" of the spectrum and correlates with the perceived brightness of sound. According to Eq. (22) from Week 8 slides:

$$SC = \frac{\sum_k \quad f_k \ \mid X(k) \mid}{\sum_k \quad \mid X(k) \mid}$$

where $f_k$are the frequency bins and $X(k)$the magnitude spectrum. Each frame's SC was averaged to form a feature vector. High SC values typically indicate energetic emotions (e.g., anger, happiness), while low SC values correspond to calmer emotions (e.g., sadness).

The SVM classifier (linear kernel, C = 0.1) trained on SC achieved **43.75% accuracy**, which aligns with the expected benchmark (~43%).

2.c Spectral Bandwidth (SBW)

SBW measures the spread of frequencies around the centroid. The tutor noted that the earlier implementation incorrectly applied a square root. The corrected version follows Eq. (23):

$$SBW = \frac{\sum_k \quad (f_k - SC)^2 \mid X(k) \mid}{\sum_k \quad \mid X(k) \mid}$$

This reflects the second central moment of the spectrum. Emotions with sharp timbral variation (anger) produce higher SBW, while smoother tones (sadness) have narrower bandwidths. Using the corrected implementation, the linear SVM achieved **50.00% accuracy**, consistent with theoretical expectations.

2.d Spectral Band Energy (SBE)

SBE quantifies energy concentration within specific Mel bands. The original implementation lacked the denominator, which was corrected per Eq. (24):

$$SBE_i = \frac{\sum_{f=L_i}^{H_i} \quad \mid X(f) \mid^2}{\sum_f \quad \mid X(f) \mid^2}$$

where $L_i$and $H_i$define the boundaries of band *i*.
The resulting SBE features better captured global energy distribution. The linear SVM achieved **59.38% accuracy**, outperforming SC and SBW.

2.e Mel Filterbank Energy Visualisation

26 Mel filters spanning the perceptual range. Each filter captures energy within a narrow band, which the SBE exploits to highlight frequency regions sensitive to emotion changes. This visual confirms that the filters overlap smoothly and maintain scale linearity in the Mel domain.

2.f Performance Summary

Feature	*Accuracy (%)*	*Remarks*
SC	43.75	Baseline; brightness-based cue
SBW	50.00	Wider frequency distribution aids discrimination
SBE	59.38	Energy-based; captures intensity variation

Among handcrafted features, **SBE > SBW > SC**, suggesting emotional cues are more correlated with energy than with centroid or spread.

3. SVM CLASSIFICATION

3.a 1D CNN on MFCC Features

A 1D CNN was trained using MFCC sequences (13 coefficients × time). The architecture comprised two convolutional layers (kernel size 3, filters 64 and 128) followed by max-pooling, dropout (0.2), and two dense layers. Dropout was reduced from 0.5 to 0.2 as suggested, allowing better gradient flow. The model trained for 60 epochs using Adam (lr = 1e-3).

Result: **Accuracy = 62.50%**, showing a clear improvement over handcrafted features due to automatic extraction of temporal dependencies in MFCCs.

3.b 2D CNN on Spectrogram Images

For the 2D CNN, log-Mel spectrograms were resized to 128 × 128 pixels. The network consisted of three convolutional blocks with ReLU activation, batch normalization, max-pooling, dropout (0.2), and two fully connected layers. Early stopping and learning-rate scheduling were used to prevent overfitting.

Training without augmentation yielded **65.63% accuracy**, the best among all methods. This improvement demonstrates the CNN's ability to learn hierarchical spectral-spatial features.

3.c Data Augmentation Discussion

The initial design applied image-style augmentations to spectrograms; however, the tutor correctly observed that these violate time-variance in speech (flipping in time reverses speech dynamics). Therefore, augmentation was limited to **time-stretching** and **noise addition**, which preserve temporal coherence. Removing unrealistic augmentations improved generalisation.

3.d Performance Comparison (Deep vs Handcrafted)

Method	*Accuracy (%)*	*Comment*
SC + SVM	43.75	Weak frequency-based cue
SBW + SVM	50.00	Moderate spread information
SBE + SVM	59.38	Energy distribution strongest
1D CNN (MFCC)	62.50	Temporal feature extraction
2D CNN (Spectrogram)	65.63	Superior spatial-spectral modelling

Deep learning models clearly outperform handcrafted features by **> 6 percentage points**, validating their higher representational power.

4. ANALYSIS AND DISCUSSION

Handcrafted features such as SC, SBW, and SBE provide interpretable spectral characteristics but lack temporal modelling. Linear SVM's simplicity limits discrimination when emotional features overlap. Conversely, CNNs learn both local and global dependencies, explaining their higher performance.

The results also highlight that energy distribution (SBE) aligns most with perceived emotion intensity, making it the most effective single handcrafted descriptor. The 2D CNN surpasses the 1D CNN because spectrograms encapsulate both time and frequency information, offering richer cues for pattern learning.

Hyperparameter tuning (lower dropout, extended epochs, and deeper layers) was crucial. Too much dropout previously hindered convergence; reducing it to 0.2 allowed better training stability and reached the expected > 60% accuracy range.

5. LIMITATIONS AND FUTURE WORK

Despite improvement, accuracy remains moderate due to limited dataset size and potential class imbalance. Further improvements could include:

- **Transfer learning** using pretrained audio models (e.g., YAMNet, VGGish).
- **Data augmentation** with pitch shifting and time-masking that preserve emotion.
- **Recurrent or attention modules** to capture long-term context.
- **Hyperparameter search** (e.g., Optuna) for optimal architecture tuning.

Additionally, feature-level fusion could integrate interpretability and performance.

6. CONCLUSION

This task demonstrated that accurate Speech Emotion Recognition requires both robust features and effective learning models. Correcting SC, SBW, and SBE implementations significantly improved classical SVM results to expected benchmarks (43–59%). CNN-based models achieved superior accuracy (> 60%) after architectural tuning and realistic augmentation. Overall, deep learning surpasses handcrafted methods by automatically discovering emotion-salient spectral-temporal patterns.